5 Application Areas

Commercial and Industrial Applications

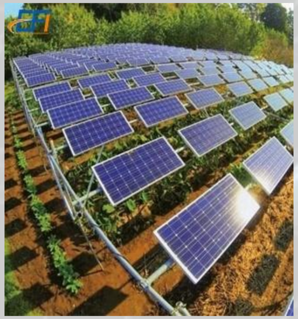
- To maximizes power density on large warehouse and to provide stable, daytime energy for corporate grids

Smart Agriculture and Farming

- To allow farm machinery to pass underneath safely

Remote Power Systems

- To power remote villages and telecom towers and to optimize battery charging cycle efficiency



6 Conclusion

- This project presents a single-axis solar tracking system with microcontroller, and models the sun's path (horizontal) to receive the maximum energy harvesting.
- In the operation, the motor is moved depending on the values of LDRs and this process is continuously repeated to adjust the panel direction toward the Sun in order to get maximum sunlight.

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MINISTRY OF SCIENCE AND TECHNOLOGY

TECHNOLOGICAL UNIVERSITY (MANDALAY)

DEPARTMENT OF MECHATRONIC ENGINEERING

PROJECT SHOW & JOB FAIR

Single-axis Solar Tracking System Using Microcontroller

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2

Aim and Objectives

The aim of this project is to construct a single-axis solar tracking system that maximizes solar energy absorption, and analyze the collected voltage output data of the system throughout a day.

The objectives are following;

- To use Light Dependent Resistors (LDRs) to detect the direction and intensity of sunlight for real-time sun-tracking.
- To employ servo motor for panel movement based on sunlight direction.
- To develop a prototype using the Arduino microcontroller.
- To evaluate the system's stability under different environmental conditions.
- To analyze the outputs of the system under varying weather conditions.

Block Diagram

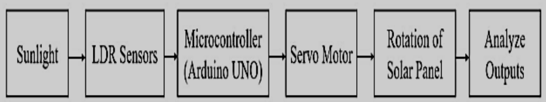

Figure 1.1. Overall block diagram of the solar tracker

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Operation

- The panel will be in default position when the system starts by turning on the switch.
- The values from the two Light Dependent Resistors, put on the side of the panel, are read.
- If the LDR1 has higher light intensity, servo motor rotates the panel to the East to track the sun's movement, and if the LDR2 detects higher, the motor rotates the panel to the West.
- The system continuously repeats this process, adjusting the panel based on the LDR readings to ensure maximum sunlight is captured.
- The voltage outputs of the system are analyzed.

Flowchart

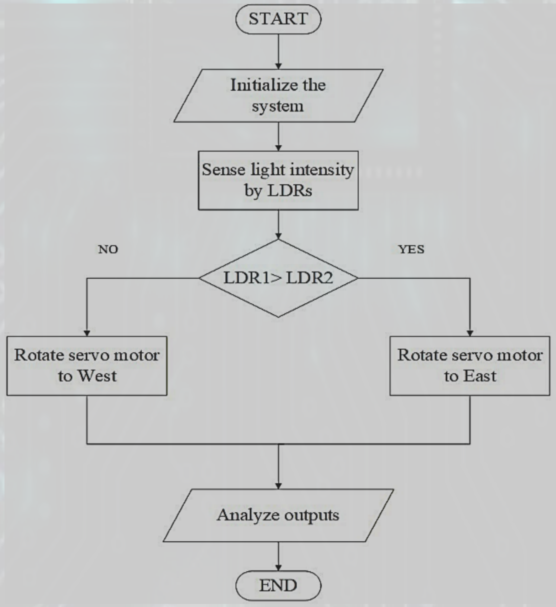

Figure 1.2. Flowchart of the solar tracker

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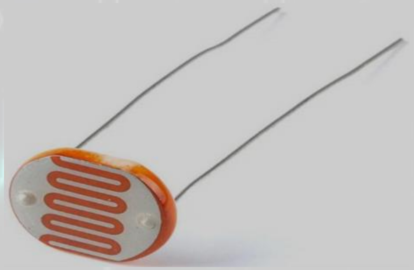
Core Components

Arduino UNO

Solar Panel

Servo Motor (MG995)



LDR Sensor